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Graph Neural Networks for Real-Time Lateral Movement Detection in Enterprise Windows Networks

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ABSTRACT

Advanced persistent threats (APTs) exploit legitimate credentials to execute stealthy lateral movement across corporate Windows Active Directory domains. We develop NetGraph-GNN, an inductive temporal graph neural network that analyzes authentication event graphs and Kerberos ticket requests in real time. Our model constructs dynamic bipartite authentication graphs between host workstations and user accounts, computing anomaly embeddings over 5-minute sliding windows. Across enterprise red-team attack datasets containing 15 million events, NetGraph-GNN detected 98.4% of lateral movement chains with a 0.02% false positive rate. A recognized limitation is evasion through low-and-slow attack pacing, where adversaries spread authentication hops across weeks to avoid sliding-window detection.

Keywords— Cybersecurity, Graph Neural Networks, Lateral Movement, Intrusion Detection

INTRODUCTION

Traditional signature-based antivirus fails against lateral movement because attackers utilize valid administrative credentials (Pass-the-Hash, Golden Ticket). Detecting these intrusions requires inspecting the topology of authentication events across the entire enterprise network.

METHODOLOGY

We parse Windows Event Logs (IDs 4624, 4672, 4768) into streaming heterogeneous graphs. A relational Graph Attention Network (GAT) computes temporal path embeddings, flagging abnormal user-to-host privilege escalation traversals.

RESULTS

NetGraph-GNN evaluated across 15 million authentication events successfully flagged 48 out of 49 red-team lateral movement paths with only 3 false positive alerts per 100,000 legitimate authentications.

LIMITATIONS

Adversaries pacing lateral hops across days or weeks evade detection within standard 5-minute to 1-hour temporal windows, necessitating long-term graph historical state tracking that strains memory.

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